Experiment 6

Develop a simple Calculator using XMLRPC.

Aim:

To develop a simple calculator application using XMLRPC (Remote Procedure Call) in Python implementing a client-server architecture.

Procedure:

1. Create server script 'calc_server.py' using simple XMLRPC server.
2. Define Arithmetic functions using server.register_function.
3. Register ~~ARI~~ functions using server.register_introspection_functions().
4. Start the server with server.serve_forever() on Port 8000.
5. Create client script 'calc_client.py' using xmlrpc client.

Command / Code:

- calc_server.py

```
from xmlrpc.server import
def add(a,b): return a+b
def subtract(a,b): return a-b
def multiply(a,b): return a*b
def divide(a,b)
    if b==0: raise ValueError("
    return a/b
def Power(a,b): return a**b
def modulo(a,b): return a%b
server=SimpleXMLRPCServer(("lo
    host',8000), logRequests=False)
for fn in[add, subtract, multiply,
    divide Power, modulo]:
    server.register_function()
Print("Calculator Server running
    on Port 8000")
Server.serve_forever().
```

Command / code:

```
# --- Cal-client.py ---
import xmlrpc.client

Proxy = xmlrpc.client.ServerProxy
("http://localhost:8000")

Print(f"10+5 = {Proxy.add(10,5)}")
Print(f"10-5 = {Proxy.subtract(10,5)}")
Print(f"10*5 = {Proxy.multiply(10,5)}")
Print(f"10/5 = {Proxy.divide(10,5)}")
Print(f"2^8 = {Proxy.power(2,8)}")
Print(f"10%3 = {Proxy.modulo(10,3)}")
```

6. Connect to the server URL
http://localhost:800

7. Call remote functions through the proxy object.

8. Display the results returned from the server.

Result:

The XML RPC calculator application was successfully developed. The server exposed arithmetic operations and the client called them remotely. All mathematical operations were verified correctly.